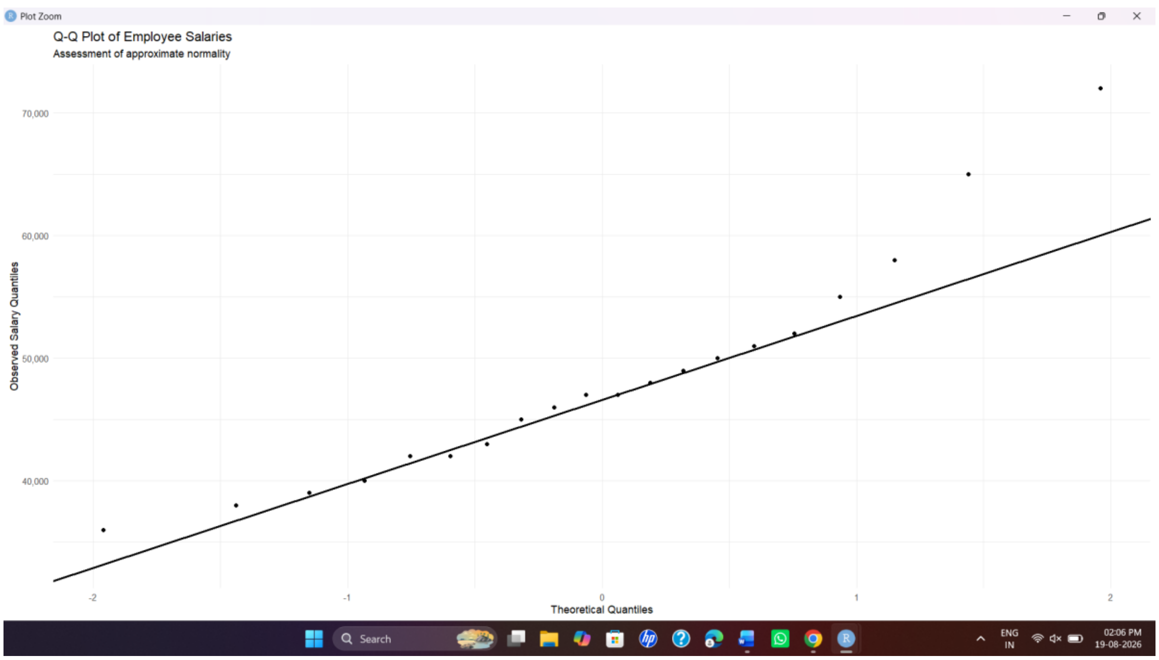
CO3-AT3

Question 1: ECDF and Q-Q Plot

```
library(ggplot2)
library(dplyr)

salary <- data.frame(
  Employee_ID = paste0("E", sprintf("%02d", 1:20)),
  Department = c(rep("IT",5), rep("HR",5), rep("Finance",5), rep("Marketing",5)),
  Salary = c(42000,45000,48000,51000,55000,
             38000,40000,42000,46000,50000,
             47000,49000,52000,58000,65000,
             36000,39000,43000,47000,72000)
)

# Original ECDF
ggplot(salary, aes(x = Salary)) +
  stat_ecdf(linewidth = 2, colour = "red") +
  labs(title = "Salary Chart", x = "Salary", y = "ECDF") +
  theme_dark()

# Redesigned ECDF
ggplot(salary, aes(x = Salary)) +
  stat_ecdf(linewidth = 1.2) +
  geom_vline(xintercept = median(salary$Salary), linetype = "dashed") +
  scale_x_continuous(labels = scales::comma,
                     breaks = seq(30000, 80000, 10000)) +
  scale_y_continuous(labels = scales::percent,
                     breaks = seq(0, 1, 0.2)) +
  labs(title = "Employee Salary Distribution",
       subtitle = "Empirical Cumulative Distribution Function",
       x = "Annual Salary (₹)",
       y = "Cumulative Proportion") +
  theme_minimal()

# Original Q-Q plot
ggplot(salary, aes(sample = Salary)) +
  stat_qq(size = 3) +
  labs(title = "Q-Q Plot") +
  theme_bw()
```

```
# Redesigned Q-Q plot
ggplot(salary, aes(sample = Salary)) +
  stat_qq() +
  stat_qq_line(linewidth = 1) +
  labs(title = "Q-Q Plot of Employee Salaries",
       subtitle = "Assessment of approximate normality",
       x = "Theoretical Quantiles",
       y = "Observed Salary Quantiles") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```

Output:



1.Original/Problematic Visualization

The original ECDF and Q-Q plot are reproduced using unclear titles, basic/poor scaling, excessive decoration, and no Q-Q reference line.

2.Redesigned Visualization

The redesigned plots use:

- Clear titles and axis labels.
- Salary values formatted as ₹.
- Percentage scale for the ECDF.
- Median reference line on the ECDF.

- Q-Q reference line.
- Minimal visual clutte

3.Brief Justification

The redesigned ECDF makes the cumulative salary distribution easier to understand, while the Q-Q plot makes departures from normality easier to identify. The improved scales and labels also make the extreme ₹72,000 salary easier to recognize.

Question 2 – Critique and Redesign Many Distributions Using Bean Plots

```
library(ggplot2)
library(beanplot)

scores <- data.frame(
  Student_ID = paste0("S", sprintf("%02d", 1:20)),
  Course = c(rep("Python",5), rep("Statistics",5),
           rep("DataViz",5), rep("AI",5)),
  Score = c(62,68,74,81,88,
          58,64,71,79,86,
          55,61,69,77,84,
          65,72,78,87,93)
)

# Original problematic visualization
par(mfrow = c(2,2))

hist(scores$Score[scores$Course == "Python"],
    main = "Python", xlab = "Score", breaks = 3)

hist(scores$Score[scores$Course == "Statistics"],
    main = "Statistics", xlab = "Score", breaks = 5)

hist(scores$Score[scores$Course == "DataViz"],
    main = "DataViz", xlab = "Score", breaks = 2)
```

```
hist(scores$Score[scores$Course == "AI"],
     main = "AI", xlab = "Score", breaks = 7)

par(mfrow = c(1,1))

# Redesigned bean plot
beanplot(
  Score ~ Course,
  data = scores,
  main = "Student Score Distributions by Course",
  xlab = "Course",
  ylab = "Score",
  ylim = c(50,100),
  what = c(1,1,1,1)
)

# Violin alternative
ggplot(scores, aes(x = Course, y = Score)) +
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_jitter(width = 0.08, size = 2) +
  stat_summary(fun = median, geom = "point", size = 3) +
  scale_y_continuous(limits = c(50,100),
                     breaks = seq(50,100,10)) +
  labs(title = "Student Score Distribution by Course",
       x = "Course", y = "Score") +
  theme_minimal()
```

2.Redesigned Visualization

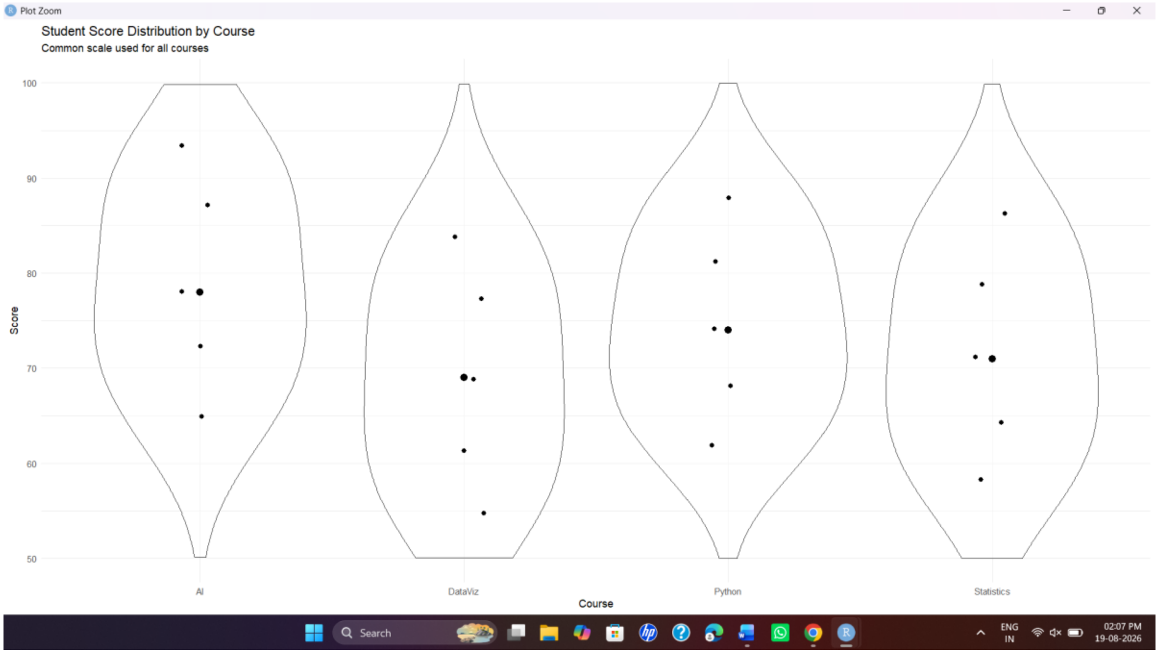
The redesign uses a **bean plot/violin plot** with:

- One common score scale from 50 to 100.
- Clear course labels.
- Individual observations where appropriate.
- Distribution shape shown for all courses in one visualization.
- Minimal visual clutter.

3. Brief Justification

The redesigned visualization allows all four courses to be compared using the same scale. It shows both the distribution and the location of student scores more clearly than separate histograms.

Output:



Question 3 – Critique and Redesign Proportion Visualizations

```
library(ggplot2)
library(dplyr)

purchases <- data.frame(
  Region = rep(c("North","South","East","West"), each = 3),
  Product = rep(c("Laptop","Phone","Tablet"), 4),
  Purchases = c(25,35,20, 30,40,25, 28,32,18, 22,38,27)
)

# Original problematic pie charts
par(mfrow = c(2,2))

regions <- unique(purchases$Region)

for(i in 1:4) {
  x <- purchases[purchases$Region == regions[i],]
  pie(x$Purchases,
      labels = paste(x$Product, x$Purchases),
      main = paste(regions[i], "3-D Pie"),
```

```
     col = c("skyblue","orange","lightgreen"),
     border = "white")
}

par(mfrow = c(1,1))

# Redesigned side-by-side bar chart
ggplot(purchases, aes(x = Region, y = Purchases, fill = Product)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  labs(title = "Product Purchases by Region",
       x = "Region", y = "Number of Purchases",
       fill = "Product") +
  theme_minimal()

# Redesigned 100% stacked bar chart
ggplot(purchases, aes(x = Region, y = Purchases, fill = Product)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Regional Product Purchase Proportions",
       x = "Region", y = "Proportion of Purchases",
       fill = "Product") +
  theme_minimal()
```

output:

2.Redesigned Visualization

Two alternatives are created:

1. **Side-by-side bar chart** – compares actual purchase counts.
2. **100% stacked bar chart** – compares proportions within each region.

Both use:

- Common scales.
- Clear labels.
- Consistent product categories.
- Minimal decoration.

3.Brief Justification

The bar charts provide a common baseline, making comparisons much easier than pie slices. The 100% stacked chart normalizes every region to 100%, so the relative product proportions can be compared directly.

Question 4 – Critique and Redesign Nested Proportions

```
library(ggplot2)
library(dplyr)
library(treemapify)
library(plotly)

sales <- data.frame(
  Region = rep(c("North","South","East","West"), each = 4),
  Category = rep(c("Electronics","Electronics","Furniture","Furniture"), 4),
  Subcategory = rep(c("Laptop","Phone","Chair","Desk"), 4),
  Sales = c(52000,38000,22000,18000,
            46000,42000,25000,21000,
            58000,35000,19000,23000,
            49000,44000,27000,20000)
)

# Original treemap
ggplot(sales, aes(area = Sales, fill = Category,
                  label = Subcategory, subgroup = Region)) +
  geom_treemap() +
  geom_treemap_text(colour = "white", place = "centre", grow = TRUE) +
  labs(title = "Original Region-Category-Sales Treemap") +
  theme_minimal()
```

```
# Redesigned treemap
ggplot(sales, aes(area = Sales, fill = Category,
                  label = paste(Subcategory, "\n₹", Sales),
                  subgroup = Region)) +
  geom_treemap(colour = "white") +
  geom_treemap_subgroup_border(colour = "black", size = 1.5) +
  geom_treemap_text(colour = "white", place = "centre",
                    grow = TRUE, reflow = TRUE) +
  labs(title = "Sales Hierarchy: Region → Category → Subcategory",
       subtitle = "Rectangle area represents sales") +
  theme_minimal()

# Sunburst alternative
sunburst <- sales %>%
  group_by(Region, Category, Subcategory) %>%
  summarise(Sales = sum(Sales), .groups = "drop")

plot_ly(
  data = sunburst,
  labels = ~Subcategory,
  parents = ~Category,
  values = ~Sales,
  type = "sunburst"
) %>%
  layout(title = "Sales Hierarchy Sunburst")
```

2.Redesigned Visualization

The redesigned treemap uses:

- Rectangle area to represent sales.
- Category-based color.
- White borders between areas.
- Black subgroup borders for regions.
- Clearer labels containing subcategory and sales values.
- Minimal decoration.

A **sunburst chart** is also created as an alternative to represent:

Region → Category → Subcategory

3. Brief Justification

The redesigned treemap makes the hierarchy clearer through borders and improved labels. The size of each rectangle directly represents sales, allowing management to quickly identify large and small sales segments.

The sunburst is useful for understanding the nested hierarchy, but comparisons of exact sizes are generally easier with the treemap.

Output:

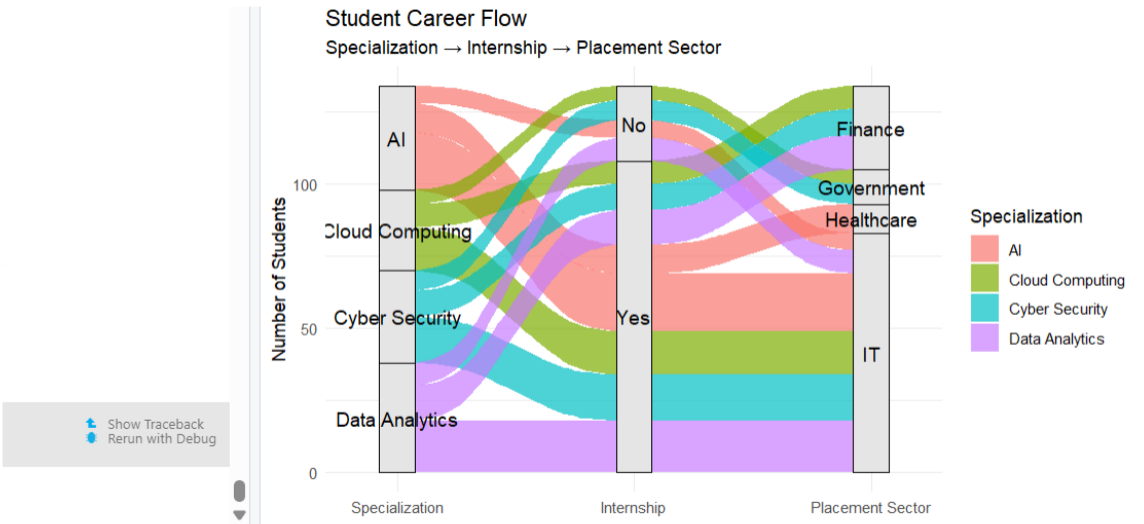
Question 5 – Critique and Redesign Flow-Based Proportions

```
library(ggplot2)
library(dplyr)
library(ggalluvial)

career <- data.frame(
  Specialization = c(
    "Data Analytics","Data Analytics","Data Analytics",
    "AI","AI","AI",
    "Cyber Security","Cyber Security","Cyber Security",
    "Cloud Computing","Cloud Computing","Cloud Computing"),
  Internship = c(
    "Yes","Yes","No","Yes","Yes","No",
    "Yes","Yes","No","Yes","Yes","No"),
  Placement_Sector = c(
```

```
    "IT","Finance","IT","IT","Healthcare","IT",
    "IT","Finance","Government","IT","Finance","Government"),
  Students = c(18,12,8,20,10,6,16,9,7,15,8,5)
)

# Original Sankey-style plot
ggplot(career,
       aes(axis1 = Specialization,
           axis2 = Internship,
           axis3 = Placement_Sector,
           y = Students)) +
  geom_alluvium(alpha = 0.4) +
  geom_stratum() +
  geom_text(stat = "stratum",
            aes(label = after_stat(stratum))) +
  scale_x_discrete(
    limits = c("Specialization","Internship","Placement Sector"),
    expand = c(.1,.1)) +
  labs(title = "Original Student Career Flow",
       y = "Number of Students") +
  theme_minimal()

# Redesigned Sankey
ggplot(career,
       aes(axis1 = Specialization,
           axis2 = Internship,
           axis3 = Placement_Sector,
           y = Students)) +
  geom_alluvium(aes(fill = Specialization), alpha = 0.7) +
  geom_stratum(width = 1/8, fill = "grey90", colour = "grey30") +
  geom_text(stat = "stratum",
            aes(label = after_stat(stratum)), size = 4) +
  scale_x_discrete(
    limits = c("Specialization","Internship","Placement Sector"),
    expand = c(.08,.08)) +
  labs(title = "Student Flow: Specialization → Internship → Placement",
       subtitle = "Flow width represents number of students",
       x = NULL, y = "Number of Students",
       fill = "Specialization") +
  theme_minimal()

# Parallel sets alternative
ggplot(career,
       aes(axis1 = Specialization,
```

```
        axis2 = Internship,
        axis3 = Placement_Sector,
        y = Students)) +
geom_parallel_sets(aes(fill = Specialization), alpha = 0.7) +
geom_parallel_sets_axes(fill = "grey90", colour = "grey30") +
geom_parallel_sets_labels(colour = "black") +
scale_x_discrete(
  limits = c("Specialization","Internship","Placement Sector")) +
labs(title = "Parallel Sets: Student Career Pathways",
     y = "Students", fill = "Specialization") +
theme_minimal()
```

Output:



2.Redesigned Visualization

The redesigned Sankey uses:

- Flow width to represent the number of students.
- Color to identify specialization.
- Clear labels for each stage.
- Stronger separation between Specialization, Internship and Placement Sector.
- Minimal visual decoration.

A **Parallel Sets** visualization is also created as an alternative.

3. Brief Justification

The redesigned Sankey makes the size of each student flow immediately visible. Wider flows represent larger numbers of students, while consistent colors help users follow each specialization through the stages.